



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc STATISTICS MAJOR COURSES

SEMESTER I

Programme	B. Sc. Statistics				
Course Code	STA1CJ101(P)/STA1MN100 (P)				
Course Title	Univariate Data Analysis				
Type of Course	Major				
Semester	I				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	HSE level Mathematics/Statistics courses				
Course Summary Objective	To make the student describe, visualize, distinguish, illustrate single variable data				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Describe types of data To understand the various types of data and emphasize the relevance of big data in statistical analysis.	U	C	Instructor-created exams / Quiz
CO2	Illustrate (numerical data) To understand and apply measures of central tendency to describe the centre of a data set	U	F	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Visualize To analyse the spread or variability within a univariate data set using measures of dispersion	U	C	Seminar Presentation / Group Tutorial Work/ Instructor-created

				exams
CO4	Illustrate (software) To provide hands-on experience applying the concepts learned in the previous modules.	U	C	Instructor-created exams / Home Assignments
CO5	To equip students with skills in effectively presenting univariate data using tables and diagrams.	Ap	P	One Minute Reflection Writing assignments/ Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Introduction to Statistics		10	10
	1	Understanding Types of Data- Categorical, Numerical Data (Discrete and Continuous)		
	2	Time Series Data, Cross-Sectional Data, Nominal and Ordinal Data		
	3	Primary and Secondary data, Design a questionnaire.		
	4	Data Sources in the Digital Age, Challenges and Opportunities in Analysing Modern Data		
	Sections from References:			
II	Measures of Central tendency		10	20
	5	Arithmetic Mean, Simple and Weighted Mean		
	6	Median, and Mode(Calculation and Interpretation).		
	7	Geometric Mean, Harmonic Mean (Calculation and Interpretation).		
	8	Comparison of Measures of Central Tendency- Scenarios for Applying Mean, Median, and Mode- Robustness of Measures, Partition values		
	Sections from References:			
III	Measures of Dispersion		15	25
	9	Absolute and relative measures of dispersion		
	10	Range, Quartile Deviation		
	11	Mean Deviation		
	12	Standard Deviation		
	13	Coefficient of Variation		
	14	Moments- Central and non-Central Moments,		
	15	Measures of Skewness based on Quartiles and Moments		
	16	Kurtosis based on Moments,		
	17	Box plot		
	Sections from References:			

IV	Introduction to R		10	15
	18	R as a calculator, R preliminaries,		
	19	Getting help, data inputting methods(direct and importing from other spread sheet applications like Excel),		
	20	Statistical software and a programming language,		
	21	Data accessing, and indexing, Graphics in R, built in functions,		
	22	Saving, Storing and Retrieving work.		
	Sections from References:			
V	Open Ended Module: Practical Applications, Case Study and Group Assignments		30	
	1	Practical exercises Hands-on using Software R: Graphical Presentation of Data, Measures of central tendency and dispersion. Case study using primary data in the form of Group Assignments and Discussions. Prepare record of at least 10 questions from Module III and IV using R Package		
	Sections from References:			
Books and References:				
Textbooks :				
1. S.C. Gupta and V. K. Kapoor, Fundamentals of Mathematical Statistics, Sultan Chand and Sons				
2. Michale J. Crawley, THE R BOOK, John Wiley & Sons, England (2009)				
References				
1. V. K. Rohatgi, An Introduction to Probability Theory and Mathematical Statistics, Wiley Eastern.				
2. Sudha G. Purohitet.al., Statistics Using R, Narosa Publishing House, , India(2008)				
3. A.M. Mood, F.A. Graybill and D C Bose, Introduction to Theory of Statistics, McGraw Hill				
4. John E Freund, Mathematical Statistics (6th edn), Pearson Edn, NewDelhi.				

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	-	3	1	2	-	-	2	-	2	-	-	-
CO 2	3	3	-	2	1	-	3	1	2	3	-	-
CO 3	1	2	-	2	-	-	2	-	2	-	-	-
CO 4	3	2	-	1	-	-	3	-	2	2	-	-
CO 5	3	2	-	-	-	-	3	-	2	-	-	-
CO 6	1	1	2	-	3	3	2	2	1	-	3	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓
CO 6	✓			